

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B.Tech. Examination

Apr-May 2018

IT248/IT201.01 Database Management System

Date: 02.05.2018, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Do as directed. [07]

- (a) What is data independence? Explain the difference between physical and logical data independence. [01]
- (b) Differentiate between the Super key and Candidate key with suitable example. [01]
- (c) Justify the statement with example: "Set Intersection is an additional relational algebra operator". [01]
- (d) Differentiate between the DDL, DML, DCL and DQL languages of database system. [02]
- (e) Consider the following two statements:- [02]
- S1: Every table with two single-valued attributes is in 1NF, 2NF, 3NF, and BCNF.
- S2: $AB \rightarrow C, D \rightarrow E, E \rightarrow C$ is a minimal cover for the set of FD's $AB \rightarrow C, D \rightarrow E, AB \rightarrow E, E \rightarrow C$. Which one of the following is CORRECT?

- A) S1 is TRUE and S2 is FALSE C) Both S1 and S2 are TRUE
- B) S1 is FALSE and S2 is TRUE D) Both S1 and S2 are FALSE

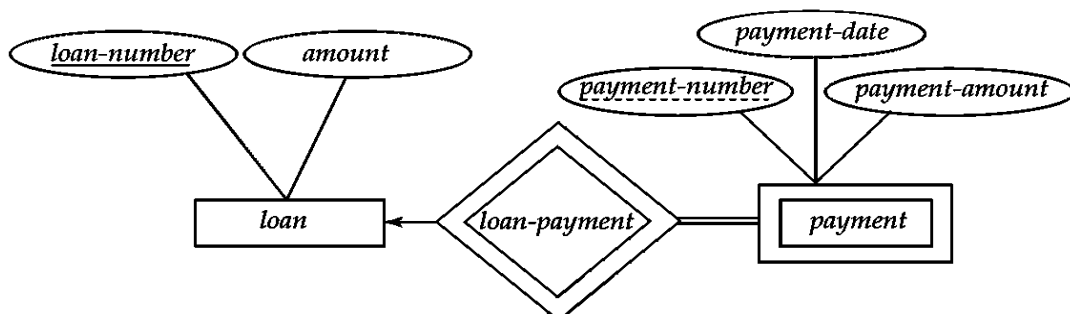
Q - 2 (a) Construct an E-R diagram for university registrar's office maintains following data. [04]
Provide appropriate relationship among entities.

- 1) course including course number, credit, syllabus
- 2) course offering including course number, year, sem, ins_id
- 3) students including std_id, name, program
- 4) instructor including ins_id, name, dept, title

Q - 2 (b) Explain the composite attribute, multivalued attribute and derived attribute in ER model with suitable example. [05]

OR

Q - 2 (b) Consider the following E-R model and explain the concept that model represent. [05]



Q - 2 (c) Explain the generalization and specialization concepts in ER model with example. [05]

OR

Q - 2 (c) Consider the following database : [05]

Customer (cust_no, cust_name, state, phone_no)

Item (item_no, item_name, price, QoH)

Invoice (invno, invdate, cust_no)

Give an expression in Relational Algebra expression for the following queries:

- Find the names of all customers who have invoice number '13'.
- Find the item number for each item having price greater than 1000 and less than 5000.
- Rename the customer relation with all its attributes.

Q - 3 (a) Consider the table below: [04]

employee						
fsname	lname	emp_id	mobile	address	sex	salary
Rohan	Chatterjee	23	19650109	731 Fondren	F	30000
Saniya	Sanyal	33	19551208	291 Voss	M	30000
Reema	Agrawal	99	19780119	561 Castle	M	25000
Rohan	Agacy	87	19801208	291 Voss	F	45000
Saniya	Sanyal	66	19801004	561 Castle	M	47000

List down all the functional dependencies present in the table and Identify the valid candidate keys of the employee relation.

Q - 3 (b) Find the canonical cover / irreducible set for following FD. And show, are they equivalent? [05]

1. $A \rightarrow B$ $AB \rightarrow C$ $D \rightarrow AC$ $D \rightarrow E$

2. $A \rightarrow BC$ $D \rightarrow AE$

OR

Q - 3 (b) Given R (ABCD) having following Functional Dependencies: [05]

$F = \{AB \rightarrow C, C \rightarrow B, BC \rightarrow D\}$. Find the highest normal form.

Q - 3 (c) You are given the below functional dependencies for relation R(A,B,C,D,E), [05]

$F = \{AB \rightarrow C, AB \rightarrow D, D \rightarrow A, BC \rightarrow D, BC \rightarrow E\}$.

- 1) Find the key attributes of above relation.
- 2) Find the highest normal form satisfies by the above relation.
- 3) Is the following dependency implied by the above set of dependencies? If so, show how using the Armstrong's Axioms: $ABC \rightarrow AE$

SECTION – II**Q - 4 Do as directed. [07]****(a)** _____ Rollback requires the system to maintain additional information about the state of all the running transactions. [01]

A Total Rollback

B Partial Rollback

C Commit

D Both (A) and (C)

(b) A schedule can be tested against conflict serializability by constructing, [01]

A Histogram

B Wait-for graph

C Precedence graph

D Bar-graph

(c) Draw a compatibility matrix for shared lock and exclusive lock. [01]**(d)** Consider a simple check pointing protocol and the following set of operations in the log. [02]

(start, T4); (write, T4, y, 2, 3); (start, T1); (commit, T4); (write, T1, z, 5, 7);

(checkpoint);

(start, T2); (write, T2, x, 1, 9); (commit, T2); (start, T3); (write, T3, z, 7, 2);

If a crash happens now and the system tries to recover using both undo and redo operations (immediate), what are the contents of the undo list and the redo list based on transactions.

(e) For a given below schedules check whether it is serializable, recoverable and not recoverable. S1:w2(a),w1(b),w1(a),r2(b),c2,c1 [02]**Q - 5 (a)** List ACID property of transaction and explain technique to implement atomicity and durability [04]**OR****Q - 5 (a)** Draw appropriate figure and explain transaction states in detail. [04]**Q - 5 (b)** Explain Three Concurrency problems and its solutions in detail [06]**Q - 5 (c)** Explain various duties performed by database administrator in detail [04]**Q - 6 (a)** Answer the following questions. [ANY TWO] [10]**i)** List and explain drawbacks of file system over database system**ii)** Justify how two phase locking (2pl) reduce concurrency.**iii)** Define Deadlock and explain deadlock prevention techniques with example**Q - 6 (b)** Differentiate Immediate v/s Deferred database recovery techniques [04]
